

KUHYAR SAEEDI

Machine Learning & Automation Engineer

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SUMMARY

M.Sc. Management Engineering student (Tor Vergata) with a software engineering background, specializing in **digital twins and predictive maintenance for industrial systems**. Four deployed ML web applications spanning manufacturing, automated optical inspection, biomedical simulation, and energy techno-economics.

EDUCATION

M.Sc. Management Engineering

University of Rome Tor Vergata · Sep 2025 - expected Sep 2027
Coursework: AI Applications in Manufacturing (**30 e lode/30**), Clean Hydrogen Technologies (**30 e lode/30**), Smart Factories (**30/30**), Digital Economy (29/30)

B.Sc. Computer Software Engineering

University of Urmia · Sep 2012 - Sep 2017

DEPLOYED PROJECTS

Manufacturing Plant Digital Twin — Predictive Maintenance [\(Python, FastAPI, Streamlit\)](#)

- Built a digital twin of a manufacturing line: equipment telemetry feeds ML models that estimate degradation and flag maintenance needs before failure
- FastAPI backend serving model outputs to a live monitoring dashboard — [live demo](#)

Fabric AOI Digital Twin — Optical Inspection [\(Python, CNN, transfer learning\)](#)

- Digital twin of an automated optical inspection station for fabric quality control, combining CNN defect detection with a simulated inspection line — [live demo](#)

Levelized Cost of Hydrogen — HRS Dashboard [\(Python, Streamlit, Excel modeling\)](#)

- Techno-economic model of on-site hydrogen refueling (alkaline electrolysis + PV): 12 configurations across Naples and Milan using PVGIS solar data — [live demo](#)
- Delivered as full academic project (report, decks, Excel model) for Clean Hydrogen Technologies — graded 30 e lode/30

Human Airways Digital Twin — Surrogate Modeling [\(Python, POD/ROM, RBF\)](#)

- Reduced-order digital twin built with surrogate modeling (POD/ROM, RBF interpolation, Latin hypercube sampling) — [live demo](#)

ADDITIONAL EXPERIENCE

Independent AI & Automation Projects [\(2022 - present\)](#)

- CNN-based industrial defect detection (ResNet18, transfer learning) and regression pipelines for electroplating coating property estimation
- NLP systems: named entity recognition, sentiment analysis, spam classification, text summarization (Transformers, PyTorch)
- Browser automation and scheduling bot: Selenium workflows for appointment monitoring with real-time Telegram notifications

TECHNICAL SKILLS

Programming: Python, R, C++ , C#

AI/ML: Deep Learning, NLP, Computer Vision, Transformers, CNNs, Transfer Learning, Predictive Maintenance, Surrogate Models (POD/ROM, RBF)

Libraries & Frameworks:

PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, FastAPI, Streamlit

Automation & Data: Selenium, Web Scraping, REST APIs, Workflow Automation

Tools: Git/GitHub, Jupyter, MySQL

DOMAIN KNOWLEDGE

Production & operations management: forecasting, MRP, inventory policies, lean / TPS, safety stock optimization. Techno-economic and levelized-cost modeling.

ONLINE LEARNING

100 Days of Code — Python Pro Bootcamp

Udemy

Machine Learning Engineer Path

Educative

Deep Learning Engineer Path

Educative

CERTIFICATION

TOEFL iBT: 103 (R:25 L:25 S:28 W:25)

LANGUAGES

Persian, Kurdish: Native

English: C1

French: B2

Italian: B1

German: A2

INTERESTS

Language learning — 1,877-day daily [Duolingo](#) streak